



DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)

September 10, 2025

Surname:

Name:

Student ID:

Row:

Column:

Time: 2 hours

Maximum Marks: 33

- The following exam is composed of **8 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided either outside these spaces or **without a motivation** will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices**, such as laptops, smartphones, tablets and/or similar. As well, you are not allowed to bring with you any kind of note, book, written scheme, and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above-mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above-mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.
- **Three of the points will be given on the basis on how quick you are in solving the exam.** If you finish earlier than 30 min before the end of the exam you will get 3 points, if you finish earlier than 20 min you will get 2 points and if you finish earlier than 10 min you will get 1 point (the points cannot be accumulated).
- The box on Page 10 can only be used to complete the Exercises 7, and/or 8.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Time	Tot.
/ 7	/ 7	/ 2	/ 2	/ 2	/ 2	/ 4	/ 4	/ 3	/ 33

Exercise 1 (7 marks)

Compare bagging and boosting in terms of how they combine models and their impact on bias and variance.

Exercise 2 (7 marks)

In reinforcement learning, explain the exploration–exploitation trade-off and discuss how the discount factor (γ) influences an agent’s long-term behavior.

Exercise 3 (2 marks)

Consider the following Python code snippet:

```
1 X = dataset.drop(columns=['target']).values
2 y = dataset['target'].values
3
4 scaler = StandardScaler()
5 Xz = scaler.fit_transform(X)
6
7 pca10 = PCA(n_components=10)
8 Z10 = pca10.fit_transform(Xz)
9 clf10 = LogisticRegression(max_iter=2000)
10 clf10.fit(Z10, y)
11 acc10 = accuracy_score(y, clf10.predict(Z10))
12
13 pca30 = PCA(n_components=30)
14 Z30 = pca30.fit_transform(Xz)
15 clf30 = LogisticRegression(max_iter=2000)
16 clf30.fit(Z30, y)
17 acc30 = accuracy_score(y, clf30.predict(Z30))
18
19 best = (clf10, acc10, 10) if acc10 >= acc30 else (clf30, acc30, 30)
20 print(f"Chosen PCA components: {best[2]}, training accuracy: {best[1]:.3f}")
```

1. Explain line by line what the code does ([1]–[20]) and summarize its overall goal.
2. What is the purpose of lines [7] and [13]?
3. Do you find any methodological issue in this snippet?

1. The snippet standardizes features, builds two pipelines (Standardize → PCA(k) → Logistic Regression) with k=10 and k=30, fits both on the full data, compares training accuracy, and picks the one with the higher training score.
2. Purpose of [7] [13] is to set PCA dimensionality (10 vs 30). Lower n_components can reduce noise/overfitting and speed training; higher may retain more variance but risk overfitting.
3. Selecting n_components by training accuracy ([19]–[20]) is optimistically biased and not a valid model selection procedure.

Exercise 4 (2 marks)

Indicate whether the following statements about Monte Carlo and Temporal Difference are true or false. Motivate your answers.

1. TD(0) has lower bias but higher variance than Monte Carlo (MC) evaluation.
2. MC policy evaluation is well-suited for continuous, non-episodic tasks.
3. Bootstrapping in TD methods refers to the practice of updating value at the end of each episode.
4. The converged value function from TD(0) policy evaluation is biased by the initial choice of $V(s)$.

1. **FALSE.** TD(0) has *higher* bias due to bootstrapping from potentially inaccurate estimates. However, it has *lower* variance because its updates depend only on a single reward and the next state's value, not the sum of all future rewards.
2. **FALSE.** MC methods require a complete episode to calculate the observed return G_t . In a continuous (non-episodic) task, episodes never terminate, making standard MC evaluation impossible.
3. **FALSE.** Bootstrapping does mean updating an estimate based on other estimates. This update happens incrementally *after each step*, not at the end of an episode.
4. **FALSE.** For a given policy, TD(0) evaluation is guaranteed to converge to the true value function V^π regardless of the initial values, provided that standard stochastic approximation conditions on the learning rate are met.

Exercise 5 (2 marks)

Indicate whether the following statements are true or false. Justify your answers.

1. Exhaustive search is guaranteed to find the optimal feature subset, and its complexity is quadratic in the number of features.
2. Wrapper methods are considered model-agnostic because the selected features are chosen independently of the learning algorithm.
3. Filter methods rank features based on their intrinsic statistical properties without training a predictive model.
4. Embedded methods, such as Lasso, are typically less computationally intensive than wrapper methods.

1. **FALSE.** While exhaustive search is guaranteed to find the optimal feature subset, its complexity is **exponential** ($O(2^m)$), not quadratic. It must evaluate all 2^m possible subsets, making it computationally infeasible for even a moderate number of features m .
2. **FALSE.** Wrapper methods are inherently **model-dependent**. They use the performance of a specific learning algorithm (e.g., an SVM or a Decision Tree) as the objective function to evaluate and select the best subset of features for that particular model.
3. **TRUE.** Filter methods are a pre-processing step. They use statistical measures (like correlation, mutual information, or chi-squared tests) to score and rank features based on their relationship with the target variable, completely independent of any learning algorithm.
4. **TRUE.** Wrapper methods require training and evaluating a model for many different feature subsets. Embedded methods integrate the feature selection process directly into the model's training algorithm itself, making them significantly more efficient by selecting features and training the model in a single process.

Exercise 6 (2 marks)

A university wants to improve student outcomes in a challenging course by offering personalized support. They have access to historical data from previous semesters.

1. You are first asked to create a model that predicts a student's **final exam score** (a value from 0 to 100). Define the data you would need, state the type of machine learning problem this is, and suggest a suitable method to solve it.
2. Now, imagine the university has developed three different tutoring strategies (e.g., peer tutoring, online modules, faculty-led workshops) but doesn't know which is most effective. They want to create an adaptive system that assigns students to a strategy to maximize the total number of students who pass the course over the semester. Define the data you would need, state the type of machine learning problem this is, and suggest a suitable method to solve it.

1. This is a classic **supervised learning, regression** problem, as the goal is to predict a continuous numerical value (the score). We would need a dataset where each row corresponds to a student. The features (X) could include quantifiable academic indicators like *midterm exam scores*, *assignment grades*, *attendance percentage*, and *hours spent on the course platform*. The target variable (y) would be the student's *final exam score*. A suitable and straightforward method would be **Linear Regression**. Other models like KNN or Gaussian Processes could also be used.
2. This is a **Multi-Armed Bandit (MAB)** problem. The challenge is to balance exploration (trying different strategies to see how well they work) with exploitation (using the strategy that currently seems best). For each student, the system needs to record the **action** taken (which of the three strategies was assigned) and the resulting **reward** (e.g., 1 if the student passed, 0 if they failed). Suitable algorithms for this problem include Upper Confidence Bound (UCB1) or Thompson Sampling (TS).

Exercise 7 (4 marks)

Suppose we collect data for a group of patients who had a specific type of cancer. The variables are age x_1 , tumor size in cm x_2 , and the probability of cancer recurrence in the next five years t . We fit a logistic regression and produce estimated coefficients: $w_0 = -5$, $w_1 = 0.04$ and $w_2 = 0.8$.

1. Compute the probability that a 50-year-old patient with a 3 cm tumor will have a recurrence in the next five years (provide the formula and substitute the values, leave the calculations).
2. What would the tumor size for a 60-year-old patient need to be to have a 50% chance of recurrence in the next five years?
3. Based on the learned model parameters, what insights can be gained regarding the impact of each feature on the outcome?
4. If a new, potentially crucial, variable (x_3) is added to the model, is it necessary to retrain the entire model from scratch?

1. The probability of recurrence is given by the logistic function:

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + e^{-(w_0 + w_1 x_1 + w_2 x_2)}} = \frac{1}{1 + e^{-(-5 + 50 \cdot 0.04 + 3 \cdot 0.8)}} = \frac{1}{1 + e^{0.6}}$$

2. A 50% chance of recurrence occurs when the linear combination in the exponent is zero. We set the age $x_1 = 60$ and solve for the tumor size x_2 :

$$w_0 + w_1 x_1 + w_2 x_2 = 0$$

$$\rightarrow -5 + (0.04 \cdot 60) + 0.8 \cdot x_2 = 0$$

$$\rightarrow -5 + 2.4 + 0.8 \cdot x_2 = 0$$

$$\rightarrow 0.8 \cdot x_2 = 2.6$$

$$\rightarrow x_2 = 3.25 \text{ cm}$$

3. The coefficient for age ($w_1 = 0.04$) is positive, suggesting that the risk of recurrence increases with age. The coefficient for tumor size ($w_2 = 0.8$) is also positive, indicating that larger tumors are associated with a higher probability of recurrence. Both seem reasonable findings in oncology.
4. If a new variable x_3 (e.g., tumor grade) were added, the model **must be retrained from scratch**. The logistic regression loss function is not separable with respect to its features, so you cannot simply find the new coefficient w_3 while keeping w_0, w_1, w_2 fixed. All coefficients must be re-estimated simultaneously.

Exercise 8 (4 marks)

Consider a linear, hard-margin, two-class SVM classifier with parameters

$$w = [1, 3], \quad b = -2.$$

Answer the following questions providing adequate motivations.

1. How is the point $p_1 = [3, 1]$ classified according to the trained SVM?
2. Is the point $p_2 = [0, 1]$ a support vector of the given SVM?
3. Provide a point that falls on the boundary of the given SVM.
4. Let us suppose we collect new data that make the classification task not linearly separable. How would you change the SVM model to deal with the task?

1. We compute the decision function for the point p_1 :

$$w^\top \cdot p_1 + b = [1, 3] \cdot [3, 1]^\top - 2 = 3 + 3 - 2 = 4$$

Since the result $4 > 0$, the point is classified as belonging to the **positive class**.

2. We compute the decision function for the point p_2 :

$$w^\top \cdot p_2 + b = [1, 3] \cdot [0, 1]^\top - 2 = 0 + 3 - 2 = 1$$

Since the result is exactly 1, the point falls on the positive margin. Therefore, **it is a support vector** for the SVM.

3. We are looking for a point $x = [x_1, x_2]$ such that it lies on the decision boundary, meaning it satisfies the equation $w^\top \cdot x + b = 0$.

$$[1, 3] \cdot [x_1, x_2]^\top - 2 = 0 \implies x_1 + 3x_2 - 2 = 0.$$

We can choose a value for one coordinate and solve for the other. For instance, if we let $x_2 = 0$, then $x_1 - 2 = 0$, which gives $x_1 = 2$. Thus, a point on the boundary is $[2, 0]$.

4. If the data is not linearly separable, the hard-margin SVM can no longer be used. We could adapt the model in two main ways:
 - **Soft-Margin SVM:** Introduce "slack variables" that allow some data points to be misclassified or fall within the margin. This adds a penalty term for such errors to the optimization problem.
 - **Kernel Trick:** Map the data into a higher-dimensional feature space where it might become linearly separable. This is done implicitly using a kernel function (e.g., Polynomial or Radial Basis Function - RBF), which allows the SVM to find a non-linear decision boundary in the original feature space.

In either case, the model's parameters would need to be re-trained on the new data.